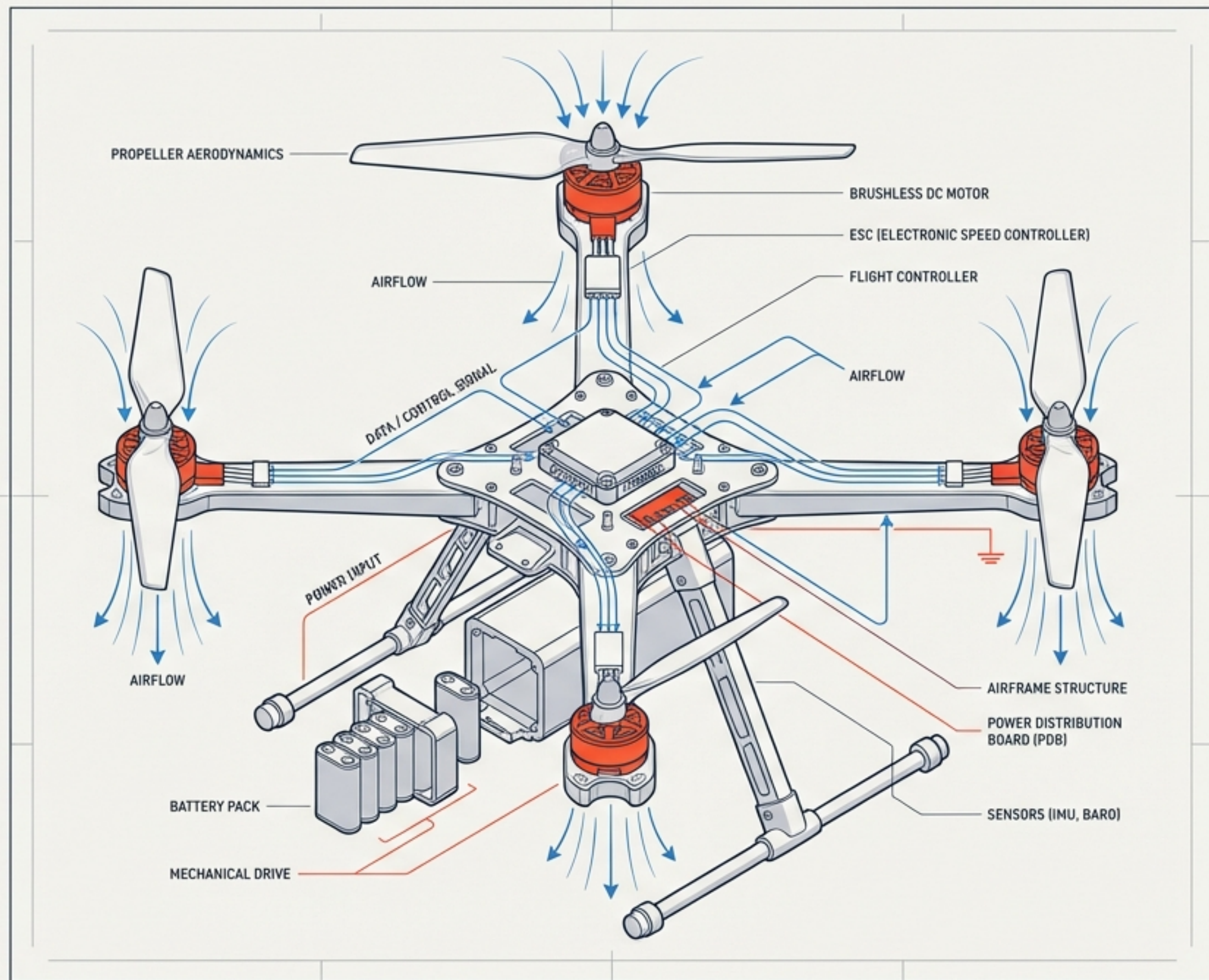


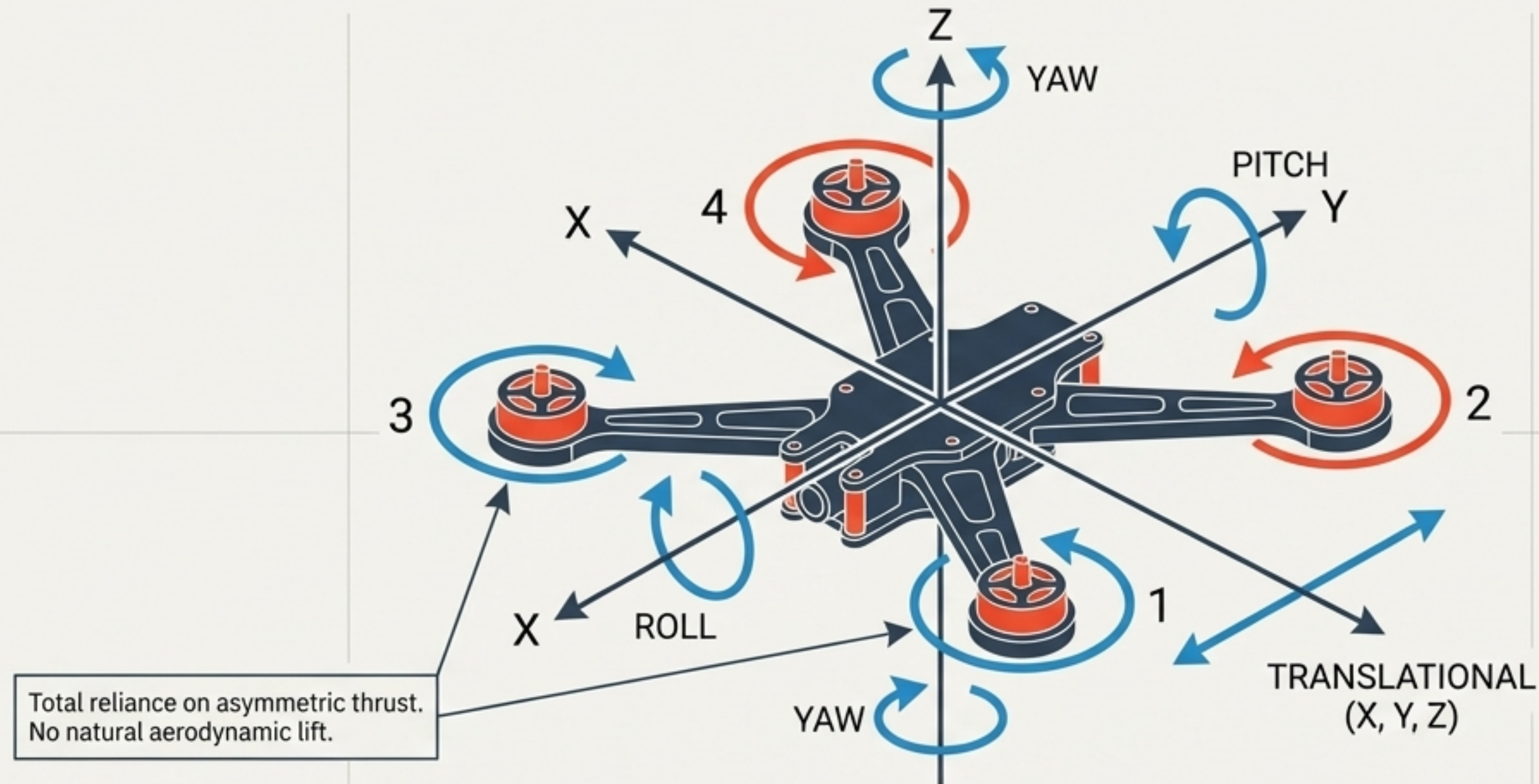
# THE ANATOMY OF FLIGHT

Deconstructing the Quadcopter Propulsion Pipeline: From Aerodynamics to Silicon





# THE INHERENT INSTABILITY OF MULTI-ROTORS



## THE CHALLENGE

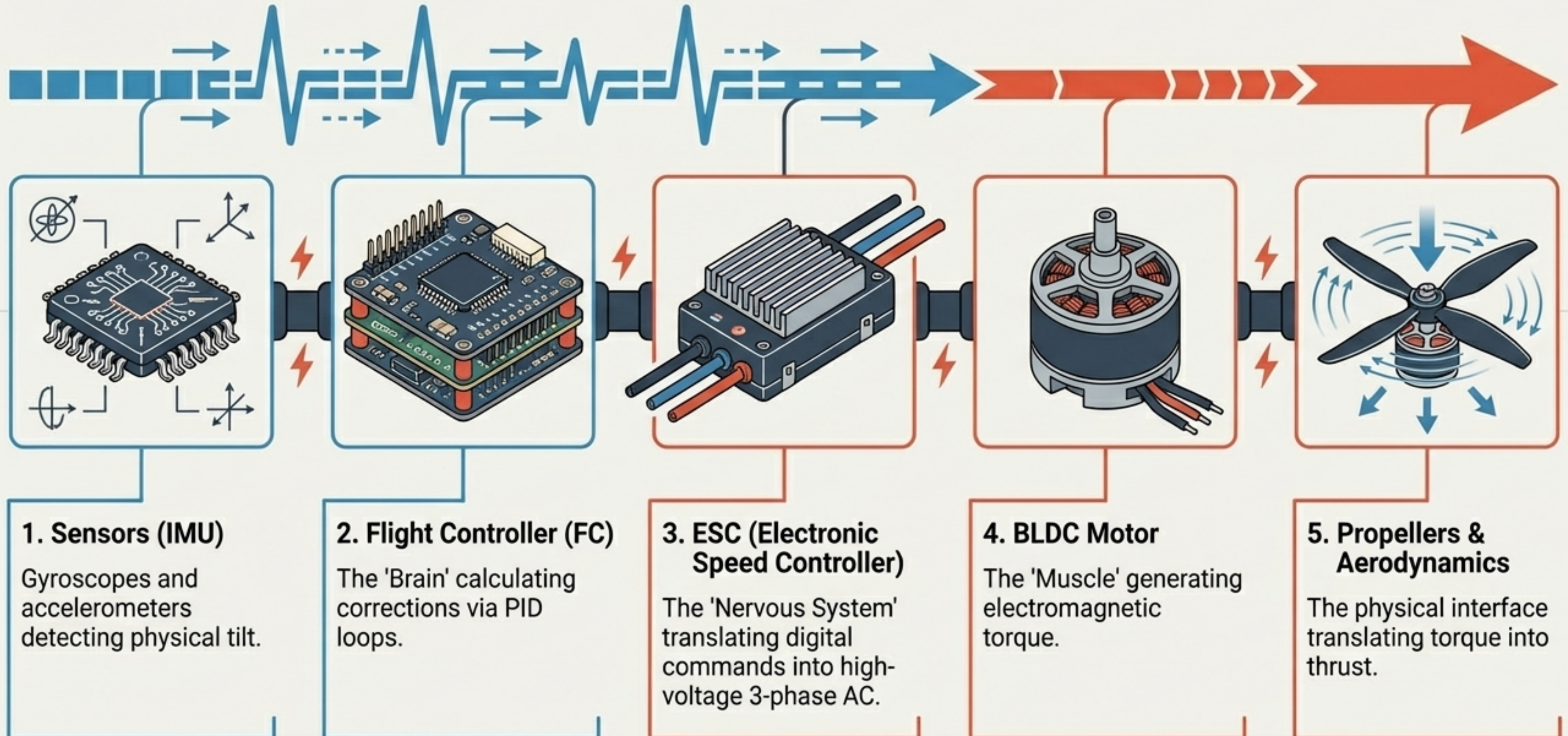
Wind gusts, motor variations, and weight shifts constantly disturb balance. Without active, microsecond-level stabilization, the craft would flip within seconds.

## THE BENEFIT

Inherent instability translates to extreme maneuverability. Stable aircraft resist turning; unstable drones respond instantly to control inputs.

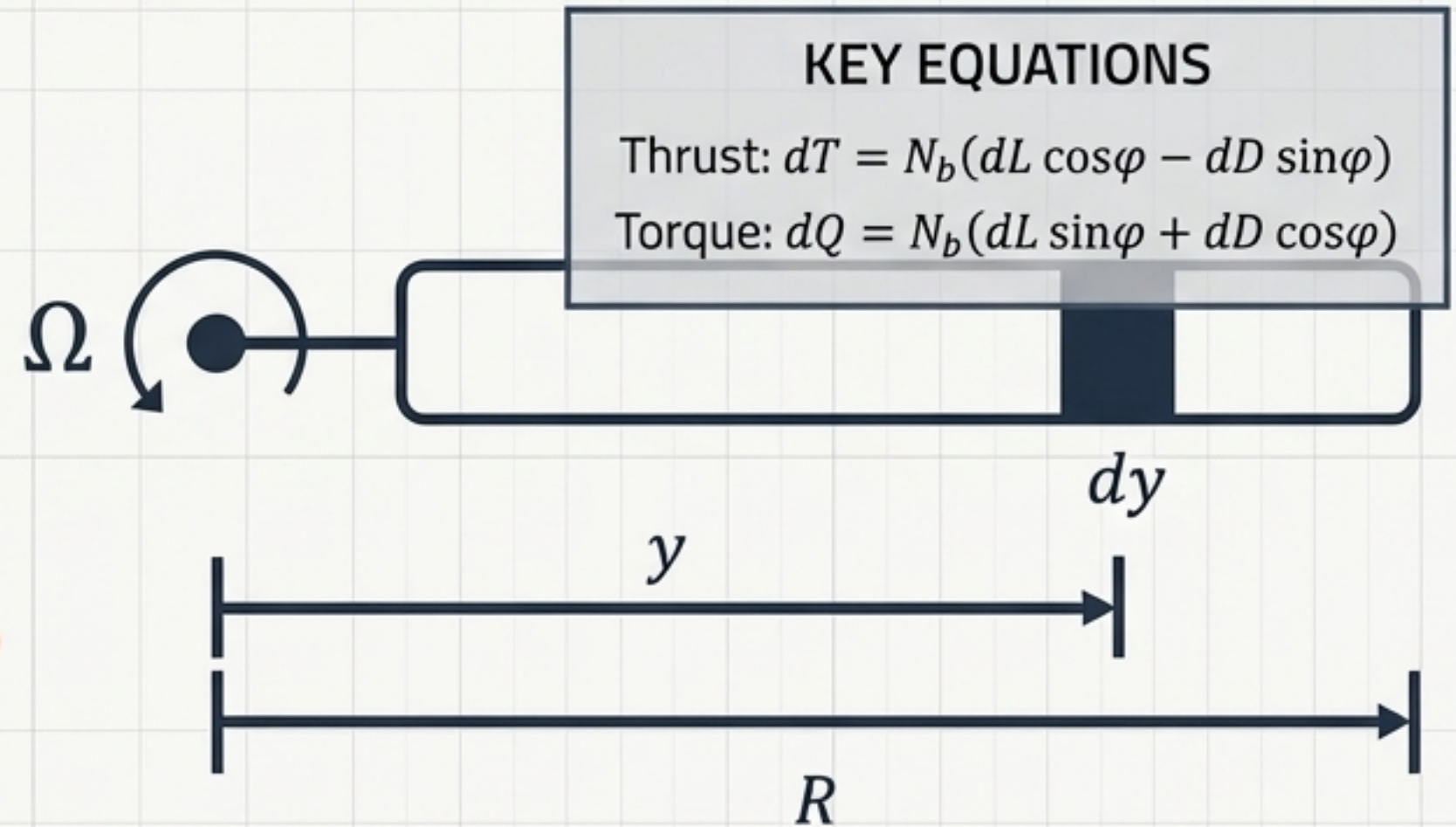
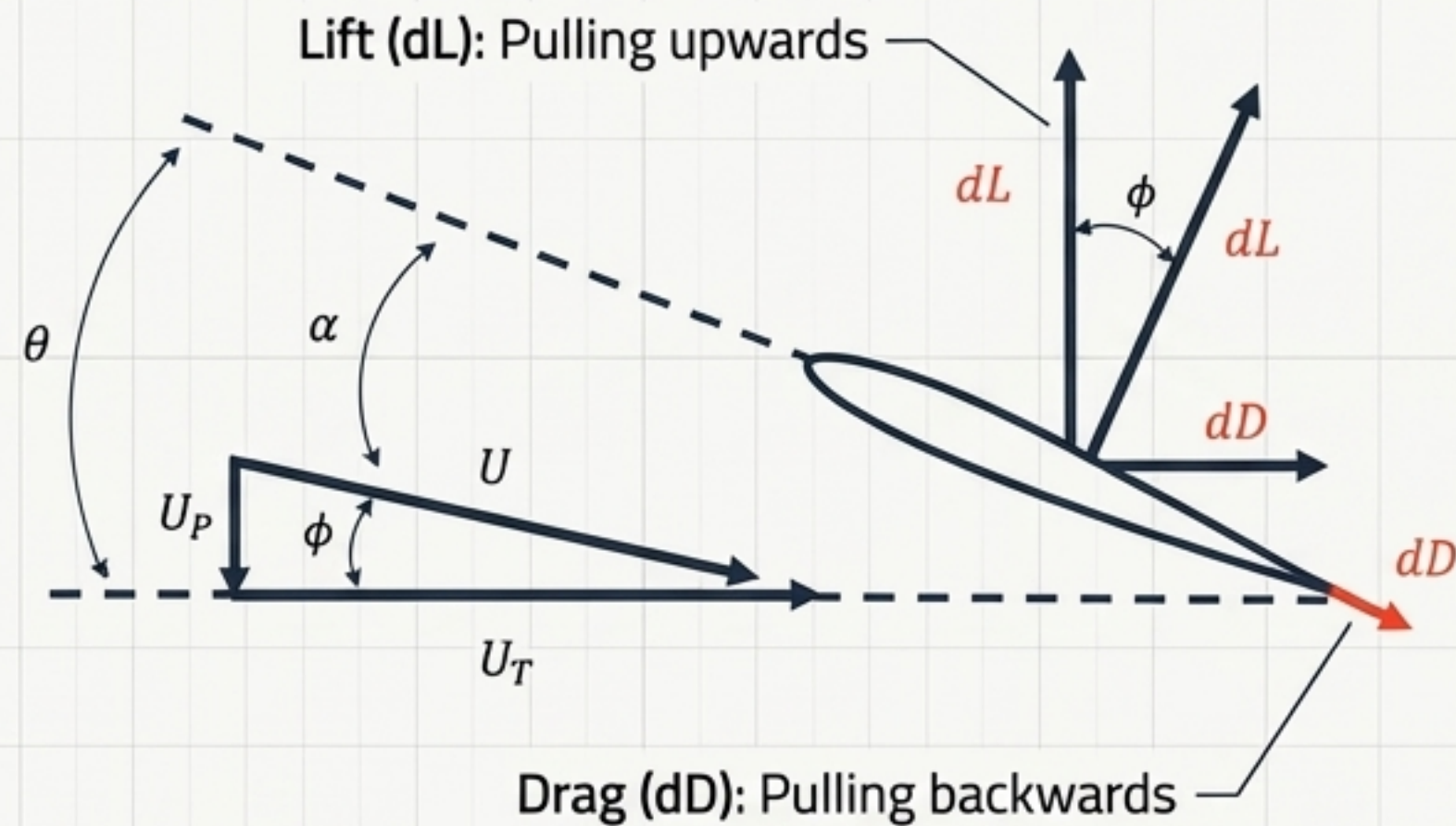


# THE PROPULSION CONTROL PIPELINE





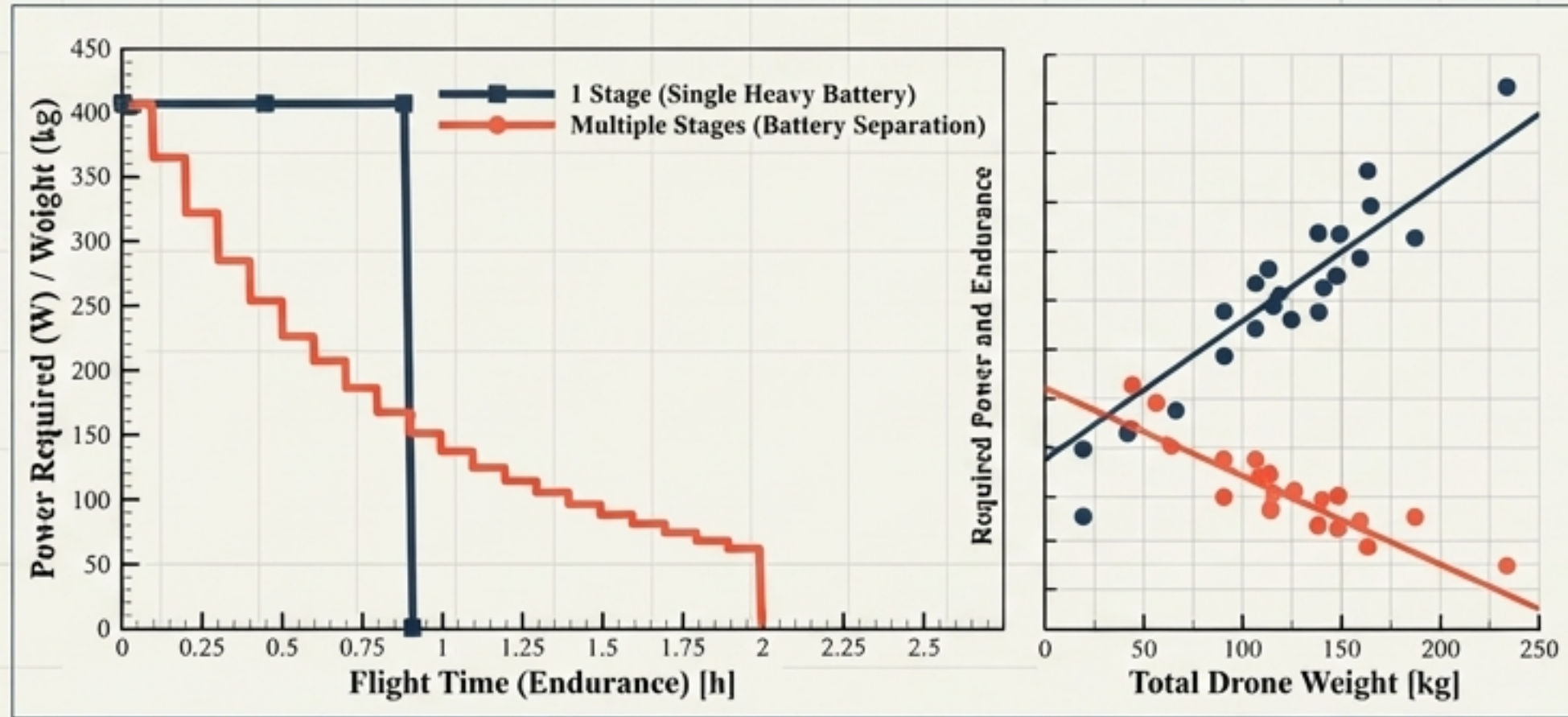
# THE AERODYNAMIC INTERFACE: BLADE ELEMENT MOMENTUM THEORY



**TAKEAWAY:** BEMT divides the propeller into geometric flow elements to predict thrust and power requirements accurately, factoring in the crucial tip-loss effect (F) where lift drops to zero at the blade's edge.



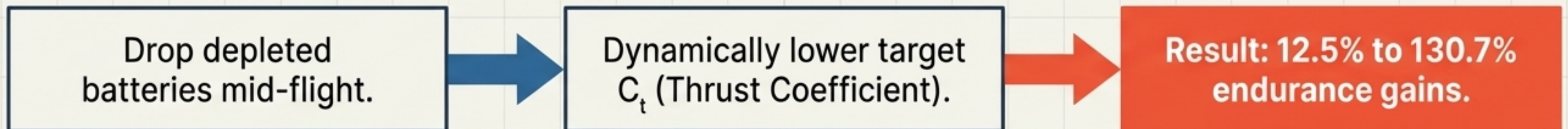
# THE WEIGHT PENALTY & THE SNOWBALL EFFECT



## THE SNOWBALL EFFECT

- Batteries consume massive proportions of takeoff weight.
- Heavier drones require more power.
- More power requires larger batteries.
- Larger batteries add more weight, demanding even more power.

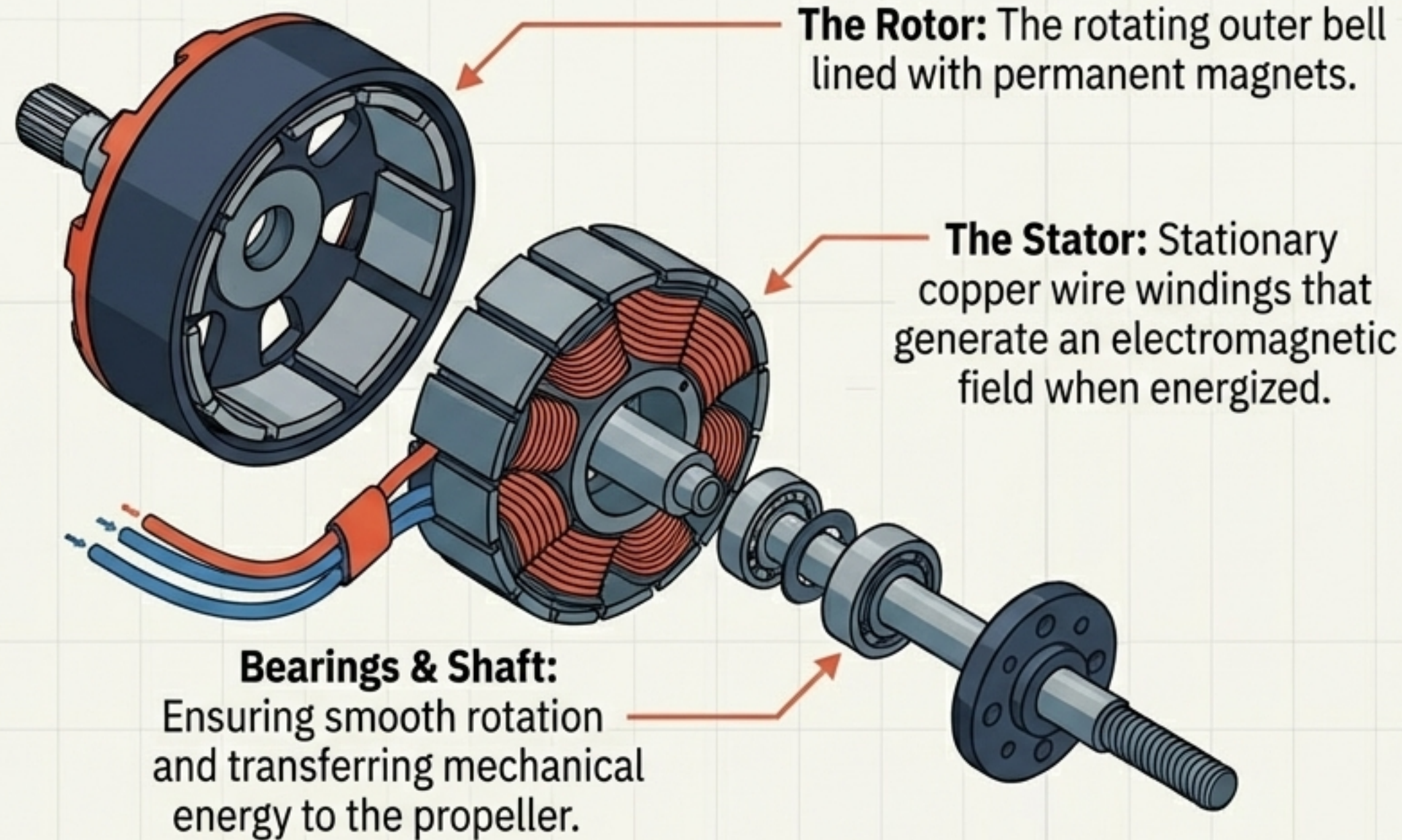
## Battery Separation Solution



(Based on ICAS commercial heavy lifter studies, e.g., DJI Matrice 300 RTK).



# THE MECHANICAL MUSCLE: ANATOMY OF A BLDC MOTOR



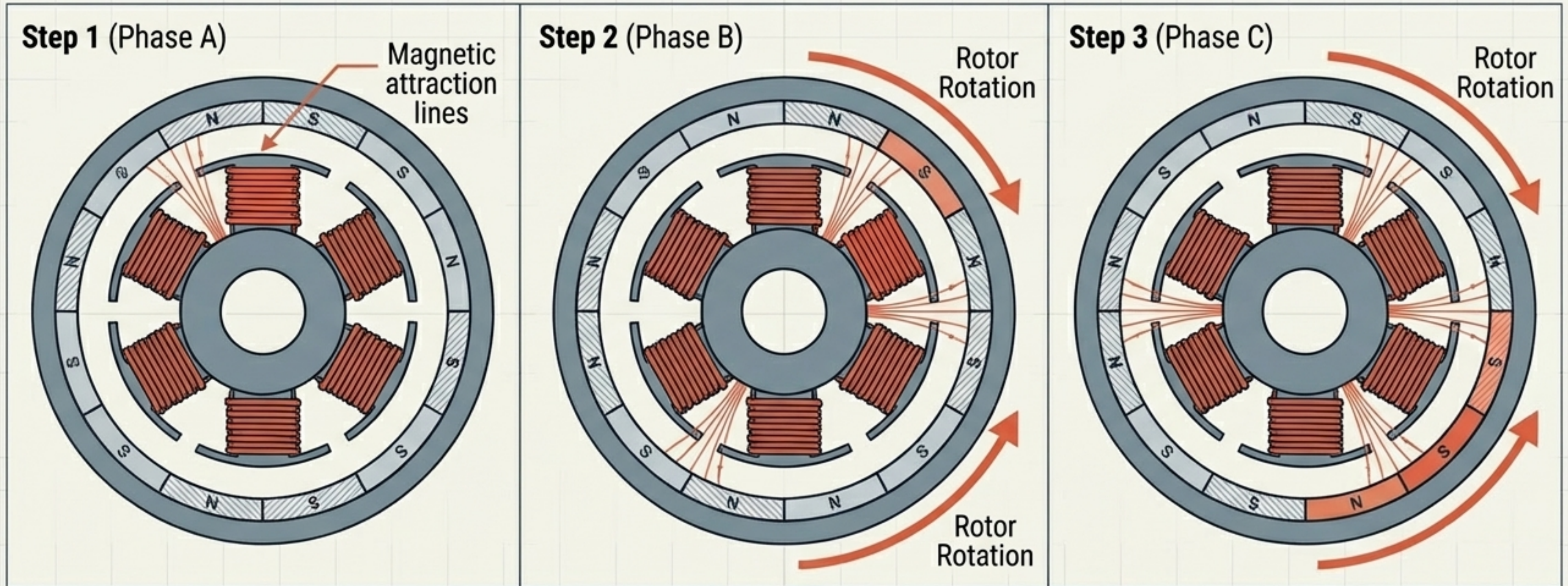
## WHY BRUSHLESS?

Unlike traditional brushed motors, BLDCs have no mechanical commutator.

There is no physical friction between moving electrical parts, resulting in vastly higher efficiency, instantaneous response times, and near-infinite lifespans.



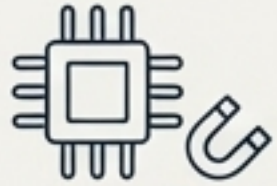
# CREATING A ROTATING MAGNETIC FIELD



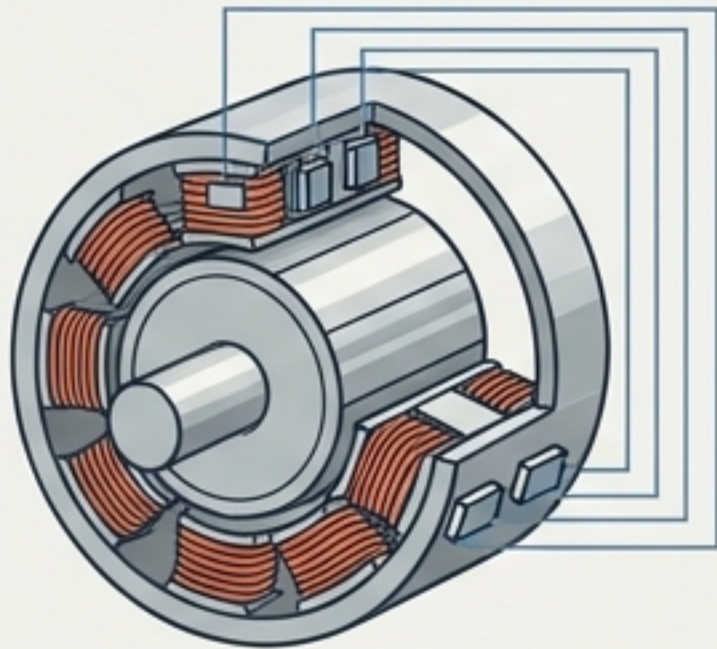
The "source of power" is the precise switching of three-phase currents. The ESC sequentially energizes Phase A -> B -> C, creating a continuously rotating magnetic field. The rotor's permanent magnets are continuously attracted and repelled, chasing the field to generate stable torque.



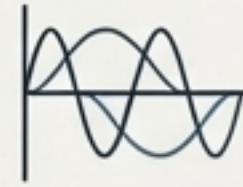
# TRACKING THE ROTOR: SENSORED VS. SENSORLESS COMMUTATION



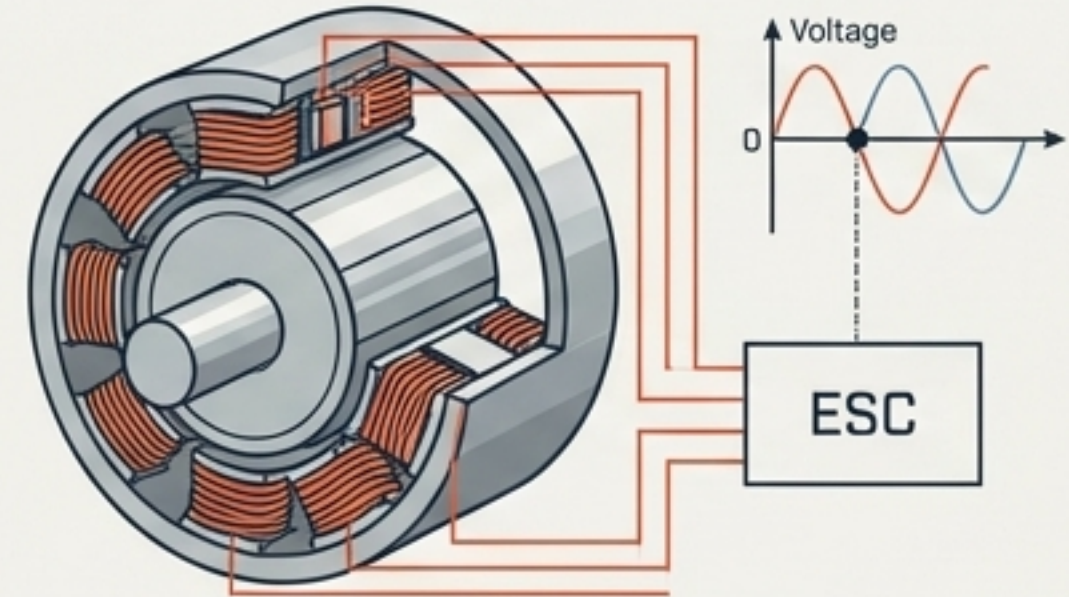
## SENSORED BLDC DRIVE



- **Mechanism:** Uses physical Hall Effect sensors to monitor rotor position.
- **Pros:** Perfect low-speed performance and smooth startups.
- **Cons:** Sensors add weight, cost, and a failure point in harsh environments.



## SENSORLESS BLDC DRIVE (UAV STANDARD)



- **Mechanism:** Detects Back-Electromotive Force (Back-EMF)—the voltage induced in the stator windings as the rotor's magnetic field cuts through them.
- **How it works:** The ESC detects the “zero-crossing point” of the Back-EMF to time the next commutation.
- **Pros:** Lightweight, highly reliable at high speeds.
- **Cons:** Difficult to read at dead stops, requiring a pre-programmed startup sequence.



# THE ELECTRONIC NERVOUS SYSTEM: ANATOMY OF THE ESC

## 1. Microcontroller (MCU)

The brain of the ESC. Houses firmware (e.g., BLHeli\_32). Interprets throttle signals and calculates motor position.

## 2. Gate Driver

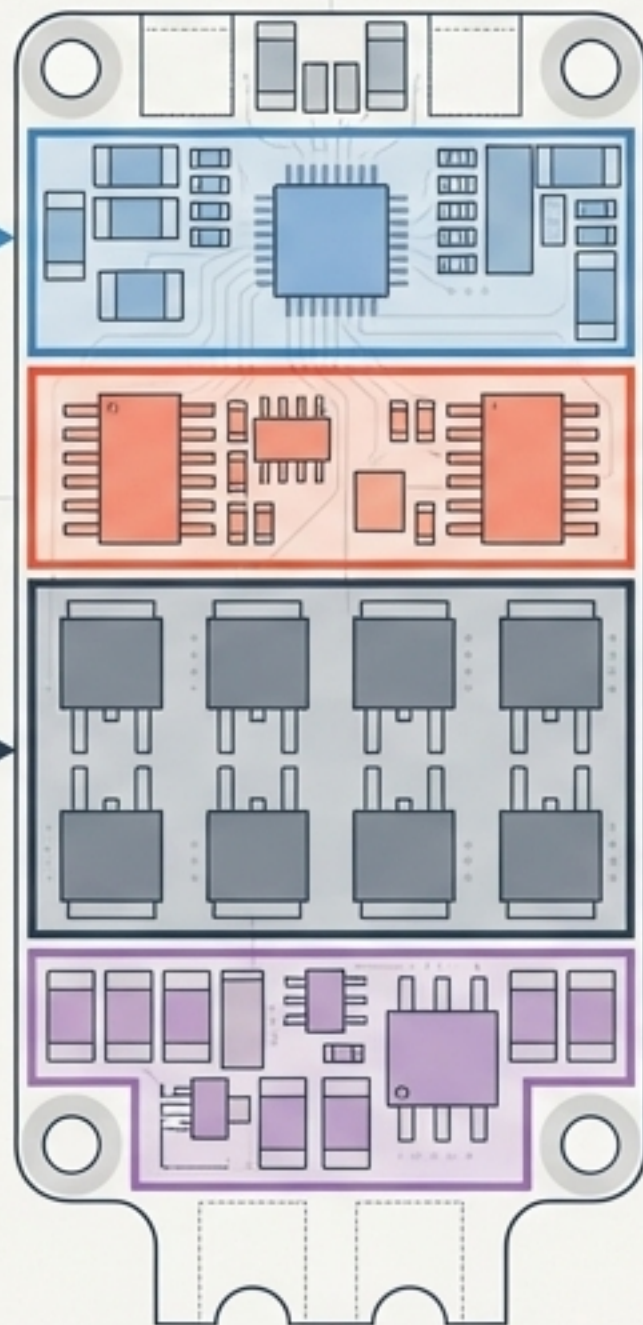
The amplifier. Takes low-voltage signals from the MCU and boosts them to high-voltage pulses.

## 3. MOSFETs

The heavy-duty switches. Six transistors that rapidly route battery power to the motor phases.

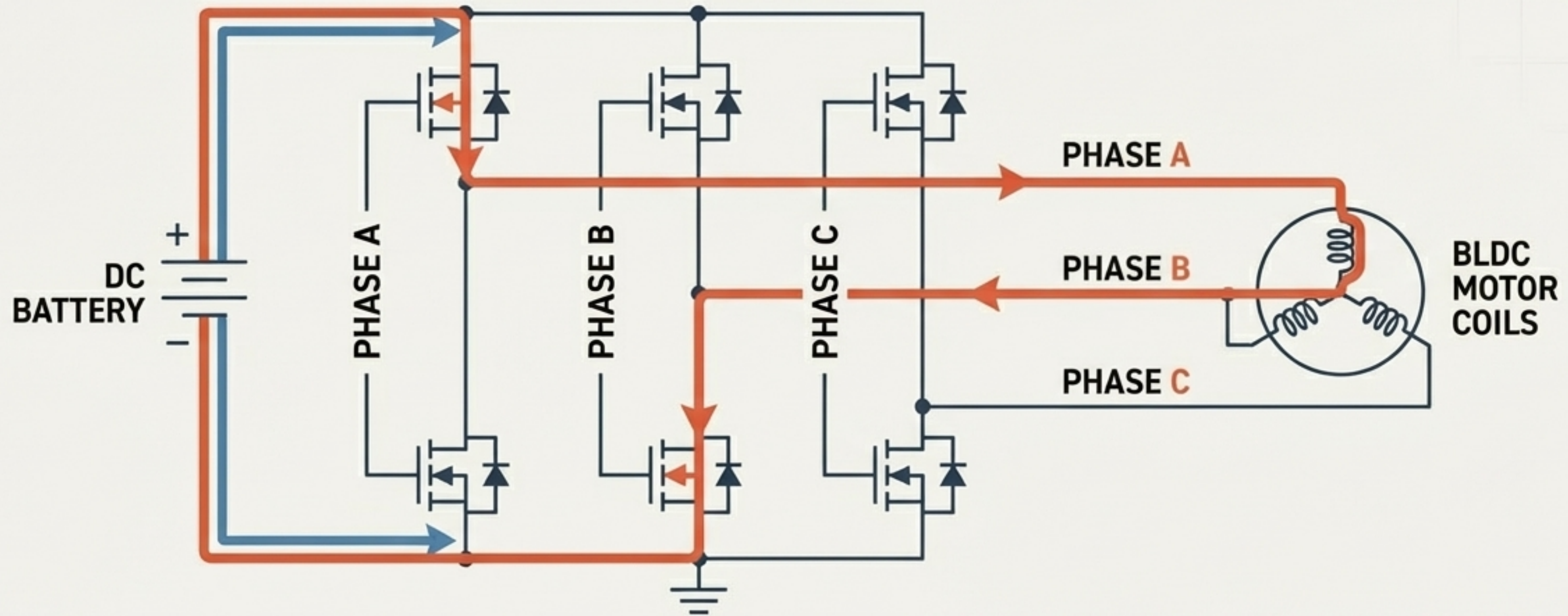
## 4. Battery Eliminator Circuit (BEC)

A voltage regulator that steps down battery power to 5V to run the flight controller and receiver.





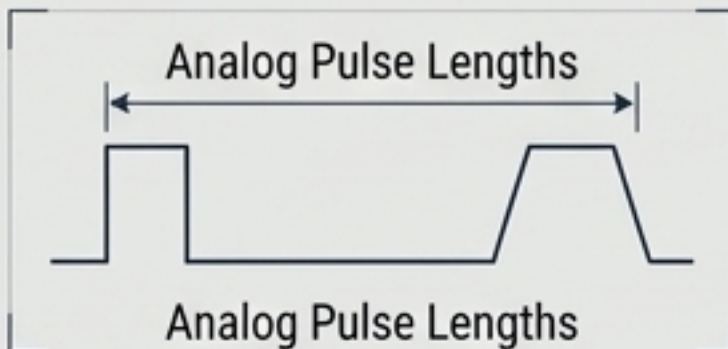
# THE 6-MOSFET BRIDGE



**ACTION FLOW:** By rapidly opening and closing these 6 switches at speeds up to 50 kHz, the ESC synthesizes a 3-phase alternating current from a direct current battery. The highlighted path shows current flowing in through **Phase A**, through the motor coils, and out through **Phase B** to ground.



# THE EVOLUTION OF ESC PROTOCOLS

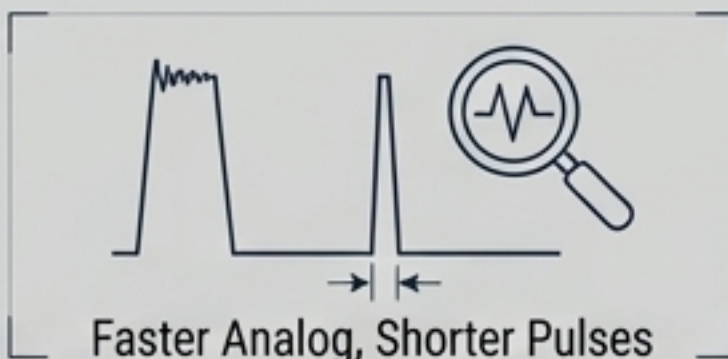
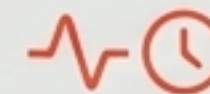


## PWM

(Pulse Width Modulation)

### Analog.

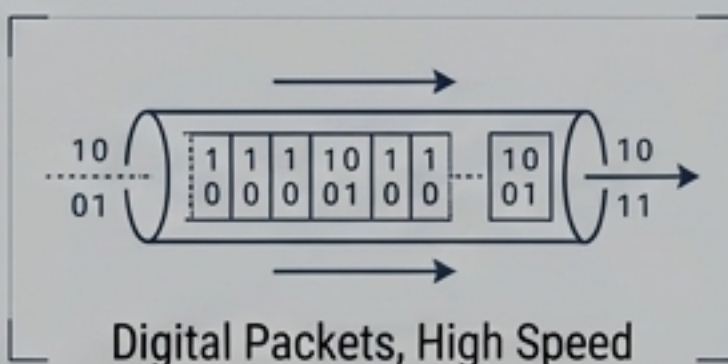
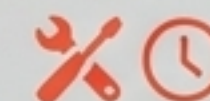
Varying pulse lengths (1000μs - 2000μs).  
50Hz to 490Hz max.  
High latency.



## Oneshot / Multishot

## Faster analog.

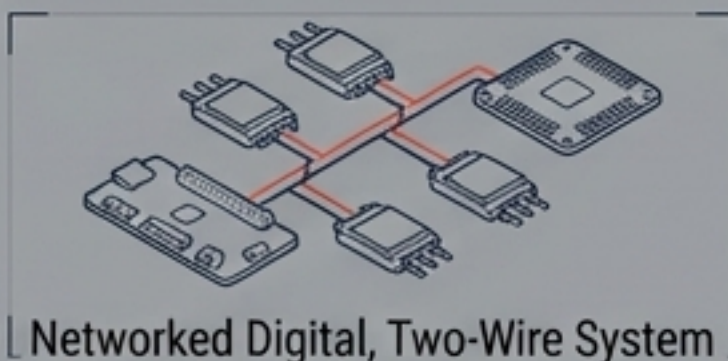
Shorter pulse lengths (down to  $25\mu\text{s}$ ).  
Requires oscillator calibration.



## DShot (Digital Shot)

## Digital packets.

Speeds up to 1,200,000 bits/sec (DShot1200)  
with a signal length of just 13μs.  
Immune to electrical noise; no calibration required.



## CAN Bus (Cyphal)

Networked digital.

Two-wire system for multiple ESCs.  
Allows priority arbitration (throttle > GPS data).





# THE TELEMETRY LOOP: BI-DIRECTIONAL DSHOT

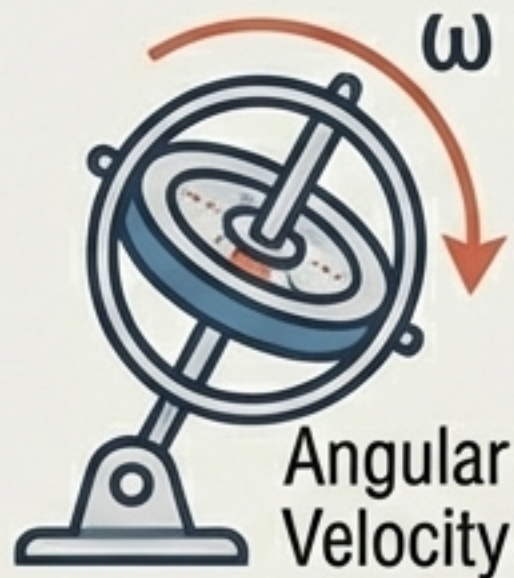


## THE PERFORMANCE BENEFIT (RPM FILTERING)

Instead of broad, latency-inducing low-pass filters that guess at motor vibration, the FC uses exact RPM data to apply dynamic 'Notch Filters'. Result: Pitch, Yaw, and Roll gyro data is scrubbed of specific motor frequencies, reducing signal delay, heat, and overall vibration.



# PERCEIVING REALITY: THE SENSOR SUITE



## GYROSCOPES

Measures angular velocity (rotation rate). Excellent for rapid, short-term movements, but accumulates minor errors over time (drift).



## ACCELEROMETERS

Measures linear acceleration and gravity ( $1G = 9.8 \text{ m/s}^2$ ). Determines absolute 'down' to correct gyro drift, but gets confused by forward acceleration.



## MAGNETOMETERS

The digital compass. Measures Earth's magnetic field for absolute Yaw/heading reference. Susceptible to electromagnetic interference from motors.

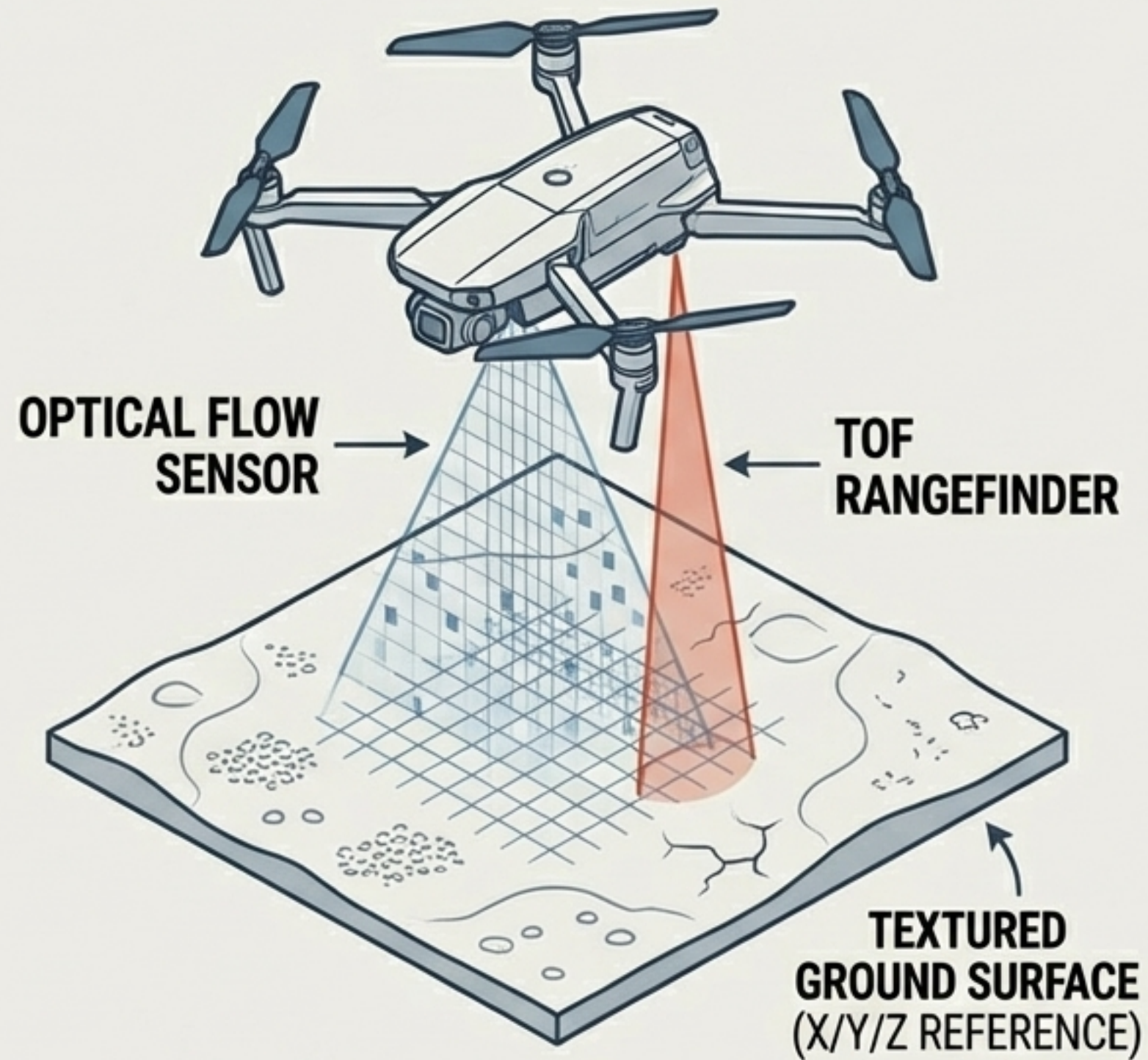


## BAROMETERS

Measures decreasing air pressure to determine altitude, enabling autonomous altitude-hold.



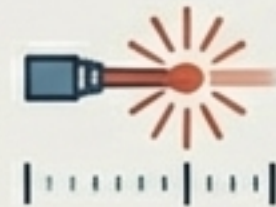
# GROUND TRACKING: OPTICAL FLOW & RANGEFINDERS



## HOW IT WORKS



- **OPTICAL FLOW:** Acts like a computer mouse sensor, taking rapid images of the ground to detect X/Y pixel movement (requires  $>60\text{Lux}$  ambient light).



- **RANGEFINDER:** Shoots lasers (pulsing at  $830\text{-}870\text{nm}$ ) to measure exact distance for Z-axis altitude hold, overcoming barometer inaccuracies near the ground.

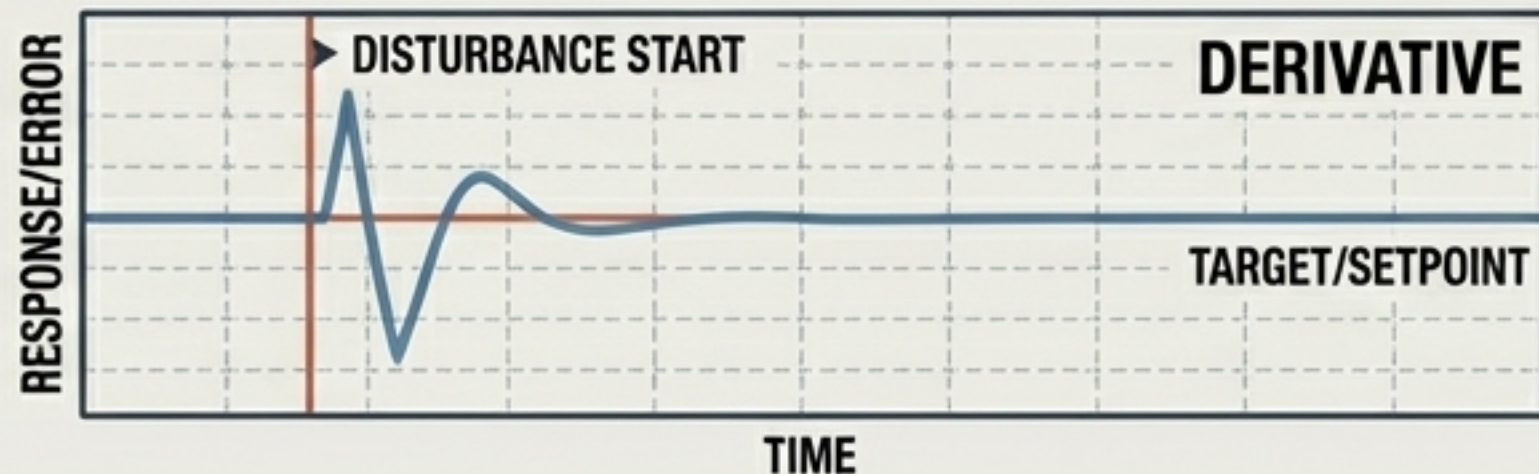
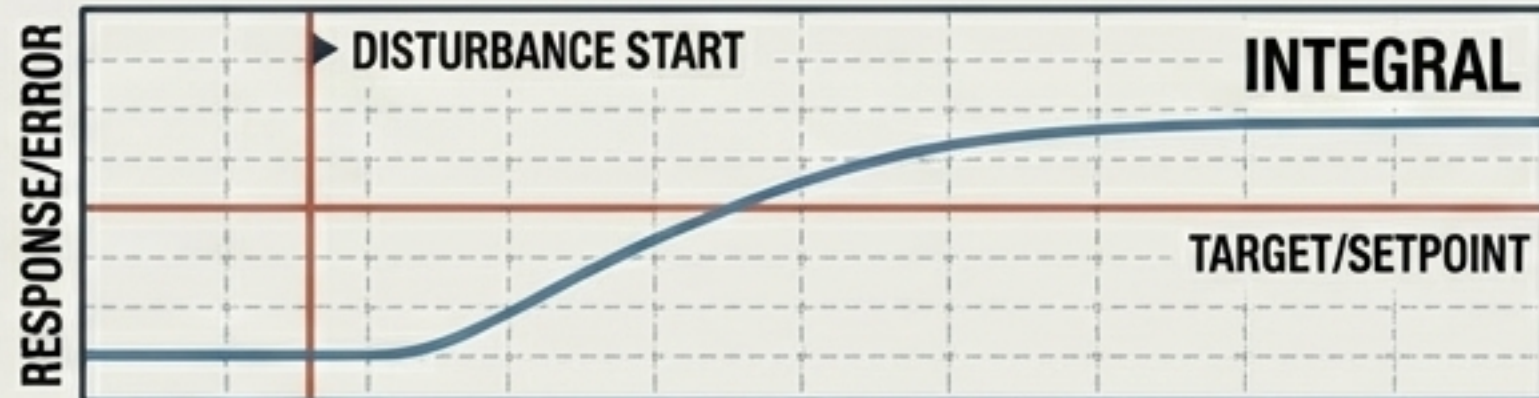
## APPLICATION



- Allows rock-solid position hold indoors or in GPS-denied environments.



# THE FLIGHT CONTROLLER: PID LOOPS



The Flight Controller processes sensor data at 500Hz to 8000Hz, running the error equation ( $e = \Omega^* - \Omega$ ) through three simultaneous calculations:



## **PROPORTIONAL (P): THE PRESENT**

Responds directly to the current magnitude of the tilt.



## **INTEGRAL (I): THE PAST**

Accumulates persistent, small errors over time to eliminate constant drift.

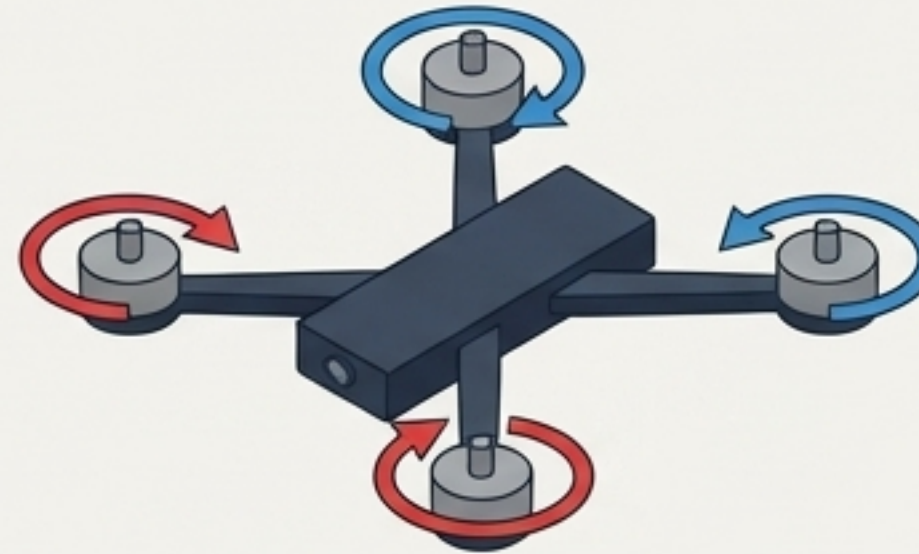
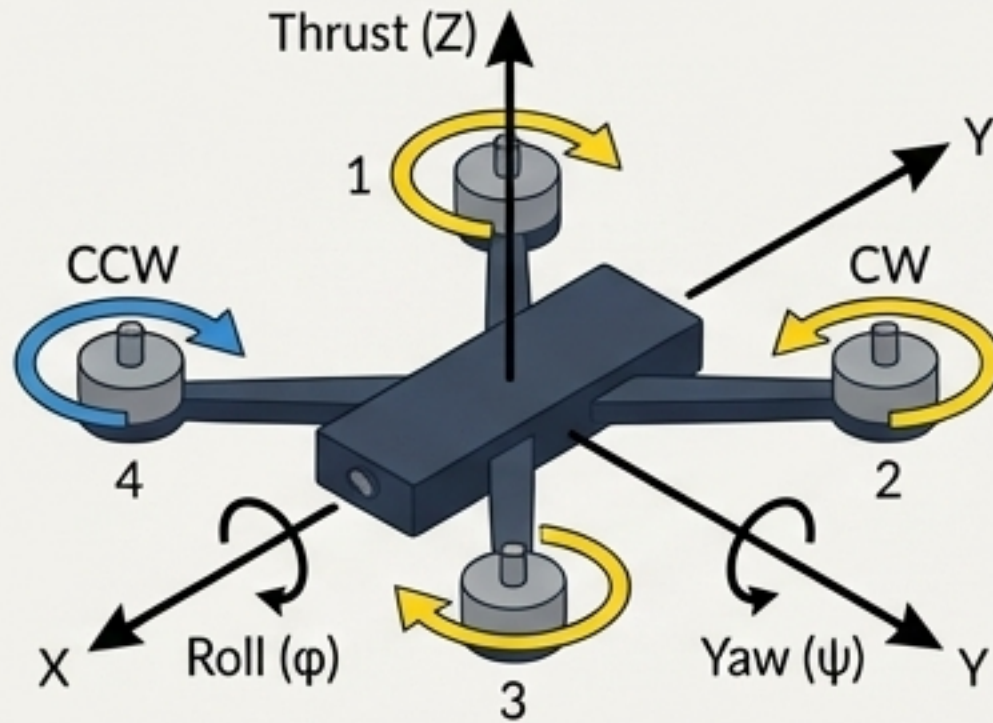


## **DERIVATIVE (D): THE FUTURE**

Looks at the rate of change. Acts as a dampener to prevent the 'P' correction from overshooting the target.



# MOTOR MIXING: TRANSLATING MATH TO MOVEMENT

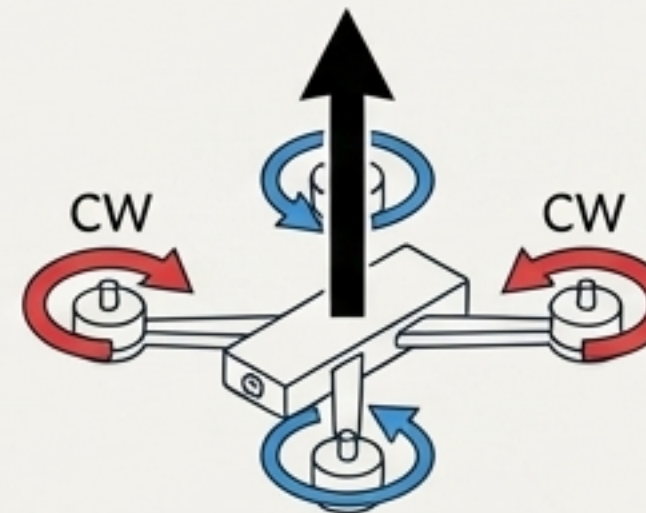
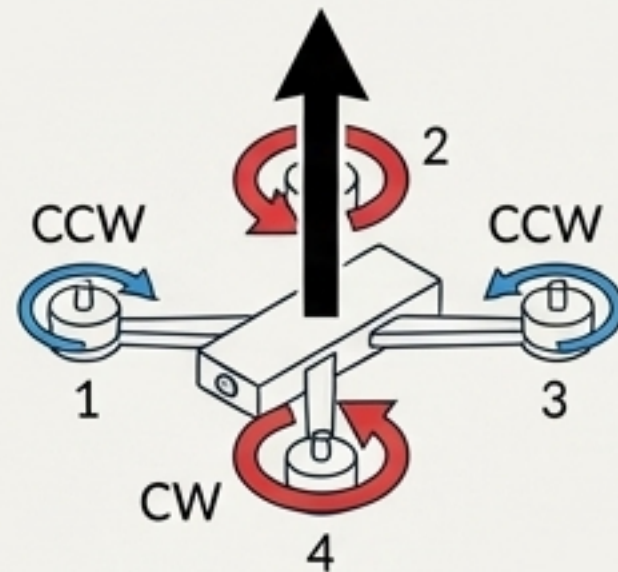
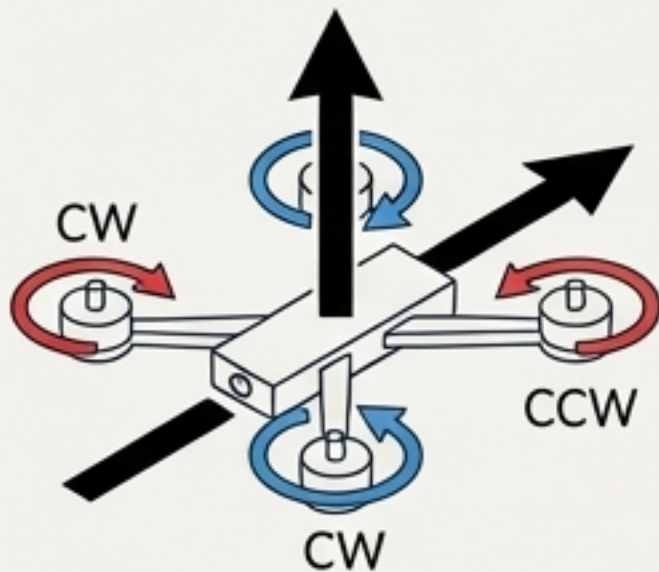
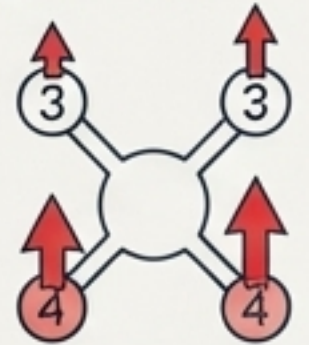


## THE MIX EQUATIONS

A central matrix translates Pitch, Roll, Yaw, and Thrust commands into discrete motor signals ( $y_i$ ).

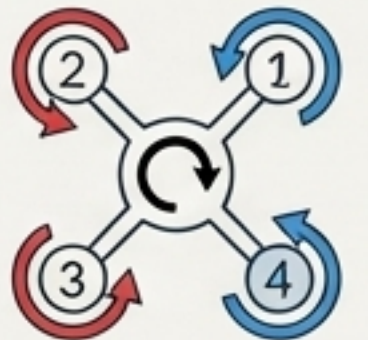
### EXAMPLE - ROLL RIGHT:

Thrust increases on motors 3 & 4 (left side).



### EXAMPLE - YAW CLOCKWISE:

No thrust difference, but a torque differential. Speed increases on the CCW motors (1 & 4), creating a net positive clockwise torque pushing the frame.

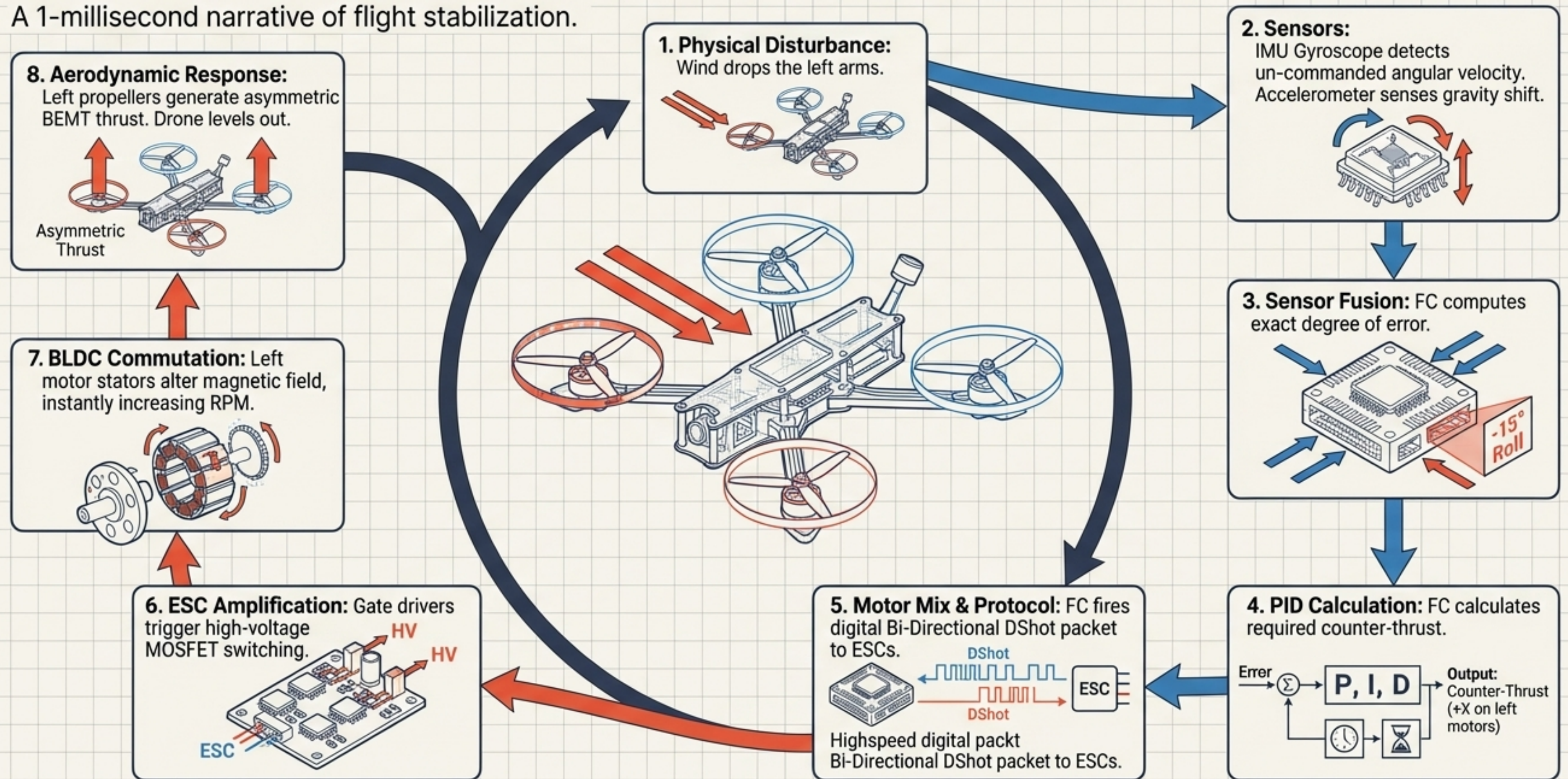


An 'X' frame quadcopter layout with CW and CCW rotation arrows.



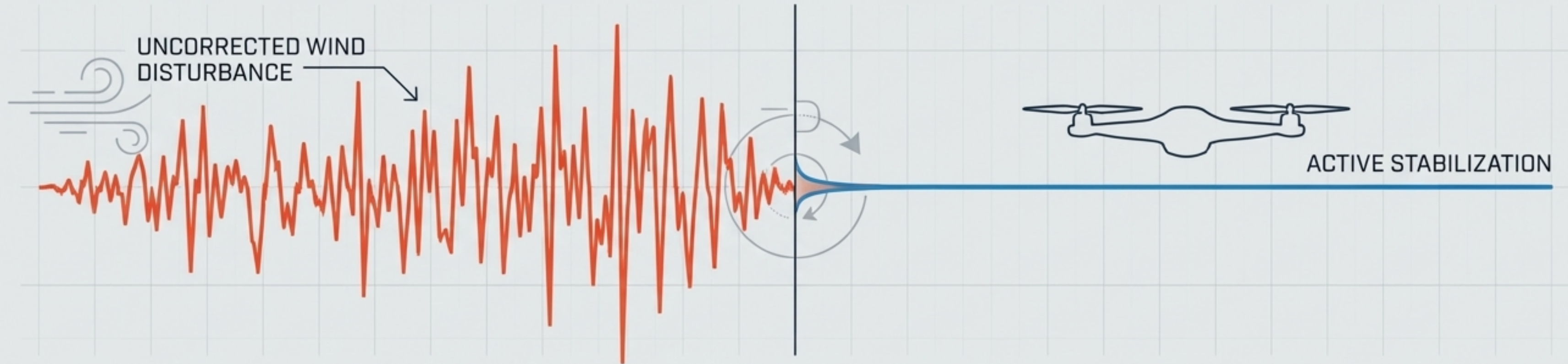
# SYNTHESIS: The 1000Hz Closed-Loop Pipeline

A 1-millisecond narrative of flight stabilization.





# CONTROLLED INSTABILITY



Drones are not inherently stable flyers; they are falling objects rescued thousands of times a second by silicon and electromagnetism.

The combination of **BEMT aerodynamics**, high-frequency **ESCs**  and precise **PID sensor fusion**   transforms a naturally chaotic system into one of the most agile and precise machines ever engineered.